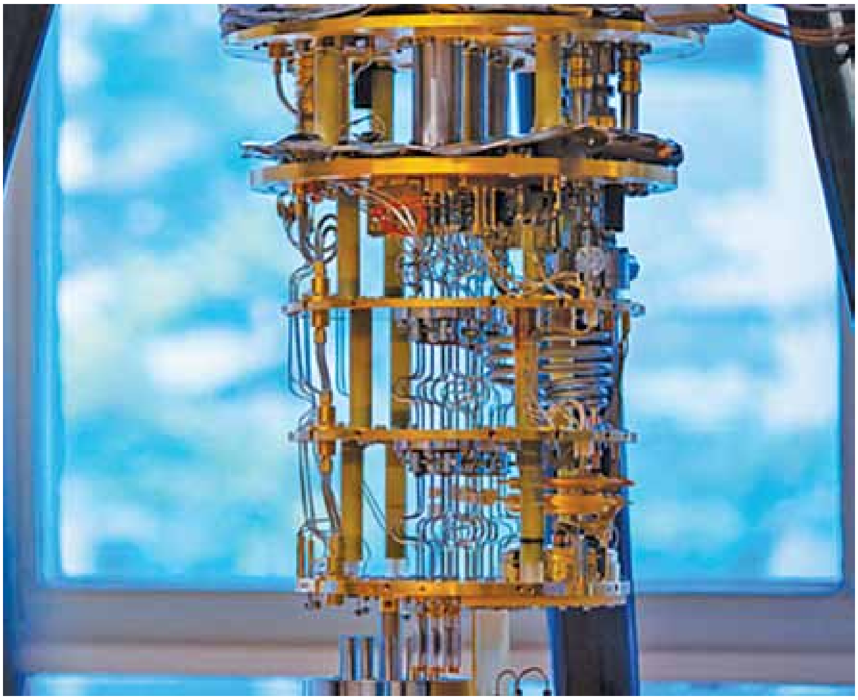

The quantum hardware where it works spin nature of the electrons (Courtesy: Intel)

sensors can detect trace amounts of pollutants, analyse gas composition, and identify biomarkers in biological samples.

These are just a few examples of the wide-ranging applications of quantum sensors. As research and development in quantum technology continue to advance, we can expect further breakthroughs and the emergence of new applications in fields such as quantum computing, quantum biology, and more. Quantum sensors hold great promise for transforming various industries and pushing the boundaries of scientific exploration.

Quantum sensors have emerged as powerful tools with immense potential across a wide range of applications. Leveraging the principles of quantum mechanics, these sensors have revolutionised metrology, enabling ultra-precise measurements in fields such as timekeeping, gravitational wave detection, and magnetic field sensing. They have paved the way for advancements in quantum communication and cryptography, offering secure transmission of information. Quantum sensors have also pushed the boundaries of imaging techniques, providing higher resolution and sensitivity in biomedical imaging, surveillance, and materials characterisation.

Additionally, they have found utility in gravity and inertial sensing, enabling precise measurements of acceleration and gravitational forces for geophysical surveys and navigation systems. Furthermore, quantum sensors contribute to the field of quantum chemistry, facilitating the accurate detection and analysis of chemical compounds. As research in quantum technology progresses, we can anticipate further breakthroughs and new applications in various fields, driving innovation and shaping the future. With their remarkable capabilities, quantum sensors hold the promise of transforming industries, advancing scientific exploration, and opening up new frontiers of knowledge. EFY